Ensemble Learning for Offline Signature Verification using Fused Deep Features

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***Abstract*— Signature verification plays an important role in document authentication. The efficient and robust system is required for forgery detection in documents in the presence of distortion such as rotation, scaling and noise. This paper proposes a writer-dependent signature verification method, where hand-crafted features are fused with one-dimensional convolutional neural network features to cater different variance in the whole verification process. Proposed feature extraction methodology is validated by bagged ensemble learning technique in which machine learning classifier's decision is finalized based on majority voting. The effectiveness of the proposed signature verification approach has been evaluated across three distinct publicly accessible datasets. The outcomes highlight the versatility of this method. Moreover, the proposed system surpasses the performance of other pre-existing systems documented in the literature.**

Keywords — signature verification, 1-d deep features, hand-crafted features, bagged ensemble learning, machine learning

I. Introduction

Biometrics is employed in many different security applications, and nowadays has gained more research interest among researchers across the globe. Biometrics can be categorized as physiological or behavioural properties. Physiological applications can be the face, iris, palm, fingerprint, and behavioral applications, such as handwriting, gait, voice, and signatures. A signature is a handwritten sample of a person's handwriting style that specifies their name, nickname, last name, or a mixture of the words used for identification, and authenticating these signatures is called signature verification. Signature verification is a crucial task in various domains, ranging from financial institutions to legal documentation and beyond. Traditionally, human experts have been responsible for determining the authenticity of signatures, a process that is time-consuming, subjective, and susceptible to errors. Signature verification processes are primarily of two types: offline and online [1].

Input images to verify offline signatures are often captured using a mobile or scanned one. To verify the online signature, a person usually uses a stylus to input their signature, which is recorded with a timestamp and signature position. In contrast to this, offline process can be catagorized either writer-independent or writer-dependent. Every time a new user's signature is added to the system, the system must be updated and retrained in a writer-dependent scenario. This makes the procedure burdensome and impractical. In an author-independent scenario, however, a generalized system that can distinguish between authentic and forged signatures without repetitive retraining must be developed. However, the introduction of machine learning techniques has brought about a substantial transformation in the field of signature verification, offering a more efficient, accurate, and automated approach.

Deep learning, a subset of machine learning, has proven to be highly effective in automatically learning and representing intricate patterns within data. When applied to signature verification, deep learning models can extract discriminative features that capture the unique characteristics of each individual's signature. This technology holds the promise of streamlining processes, reducing the risk of fraud, and increasing the overall efficiency of document authentication.

In this paper, we are presenting the fusion of hand-crafted features with the 1-d deep features with ensemble learning. Ensemble learning implies the combination of multiple individual models to design an accurate, predictive, and stronger prototype. When applied to signature verification using 1-d deep learning and hand-crafted features, ensemble technique significantly enhances the robustness and reliability of the authentication process. This approach leverages the diversity of multiple models to improve generalization, mitigate overfitting, and ultimately increase the overall accuracy of signature authentication.

The contributions of this article are:

- Hand-crafted features are fused with the 1-D Convolution Neural Network (CNN) features outperform in authenticating the signatures.
- Evaluation of the proposed methodology on publicly available diverse datasets to prove the validity of the proposed model.

This article is organized in a what that Literature review of handwritten signature verification is given in section II. Section III presents the proposed model, whereas the verification results are illustrated in section IV. The conclusion and future work are stated in section V.

II. Literature Review

Numerous strategies have been suggested and effectively executed to discern the authenticity of a signature image from whether it's been forged or not. However, an approach called dynamic time warping, which evaluates the alignment between two evolving structures, has been employed to differentiate between authentic and forged samples [3]. Furthermore, a feature extractor based on the local random transform has emerged [4]. Additionally, novel methods for verifying individual signatures have been

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introduced. These include the pixel matching technique [5], structural features derived from the signature's shape using altered direction or its refined version [6], and hybrid characteristics [7, 8], which combine local and global attributes into a visual codebook accompanied by linked attributes [9]. An eigenvalue approach on the lower and upper envelopes of signatures has also been proposed for verification [10].

Despite these techniques, most of the contemporary learning relies on feature learning representations using deep neural networks (DNNs) to authenticate signatures. In contrast to employing manually crafted features, Dey et al. introduced the SigNet network model for independent offline signature validation, utilizing a Siamese network based on CNN. Their approach involves assessing the similarities between pairs of observations, decreasing the Euclidean distance for similar pairs, and increasing the distance for dissimilar pairs [11]. Another group designed a CNN architecture with 18 layers, consisting of convolutional, pooling, rectified linear unit (ReLU), normalization, and fully connected layers. The output of one layer serves as input for the next, creating high-dimensional features and enhancing accuracy [12]. DNNs have been favored for offline signature verification due to their ability to self-learn for handwritten signature authentication.

Hafemann et al. suggested employing a deep CNN to extract features solely from input images, and this model was trained using a distinct image group [13, 14]. A Siamese network based on CNN was proposed for offline signature verification [15]. Similarly, another CNN-based solution for both writer-dependent and writer-independent signature verification was introduced the same year [16]. Transfer learning (TL) models based on GoogleNet inception V1 and V3 were also utilized on the Grupo de Procesado Digital de Seales (GPDS) dataset in the same year [17]. In 2020, Siamese networks were applied to three distinct datasets - GPDS, MCYT 75, and BHsig260 - alongside statistical measures to enhance model performance [18]. However, CNNs require minimal preprocessing and are a variant of the multilayer perceptron. In modern machine learning tasks, these computational models based on biological neurons surpass earlier forms of artificial intelligence (AI). Large-Scale Visual Recognition Challenge (LSVRC) exemplifies a large-scale task, with DL CNN methods exhibiting the highest accuracy in the ImageNet challenge [19]. Recent research on the ImageNet dataset has showcased state-of-the-art accuracy results [20].

III. Proposed Metholodology

In our paper, the fusion of hand-crafted and 1-D CNN features is presented. The proposed methodology is given in Fig. 1. Initially, the signature dataset has been divided into two parts i.e., training and testing data. The split we consider for the training and testing is 70:30 ratio. The fundamental pre-processing such as identifying the signatures on the document and image resizing has been done on the experimental dataset. After extracting the features from deep models which includes VGG16, DenseNet, MobileNet as well as hand-crafted features with 1-D CNN, it is passed to the machine learning classifier and the bagged ensemble learning technique is applied.

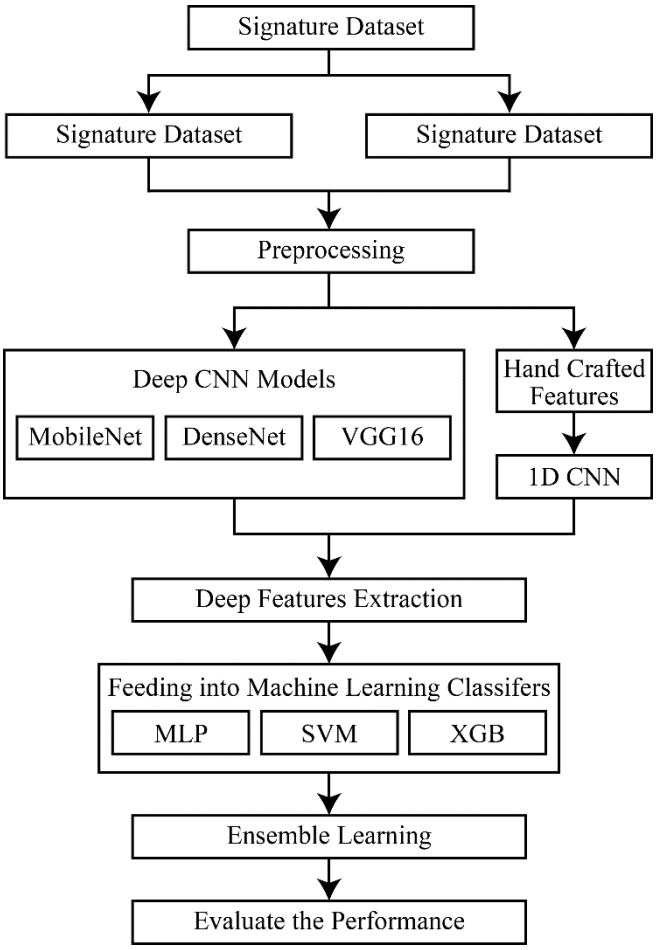


Fig.1 Proposed Methodology

A. Hand-Crafted Features

For signature authentication in digital documents, the issue of affine transformations such as rotation and scale variance are very important. The hand-crafted features are there to cater for the issues of rotation and scale invariance. These hand-crafted features are classified into structural-based, statistical-based, and model-based approaches. We are using widely used rotation invariant Local Binary Pattern (LBP), Gabor, and gray-level co-occurrence features from document textures which helps in distinguishing the different signatures. These features are concatenated for each signature sample and then forwarded to 1-D CNN architecture. Ojala et al. [21] demonstrated Local Binary Pattern (LBP) features which aim to compare neighboring grayscale image pixels. LBP has high discriminative power and gives better results in less computational time. Due to its simple implementation, it can be used for signature discrimination conveniently. Its equation is given as

$$LBP_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c)2^p, s(x) = \begin{cases} 1 & x \le 0 \\ 0 & x > 0 \end{cases} \quad (1)$$

where g_c is the given gray score of continuous central pixels, P represents the location of g_c, g_p is a gray value of a certain LBP on a spherical neighborhood of p, for all $p \in [0, p-1]$ and R is a rotating cost that represents a neighborhood area.

The Gabor filter is a pattern of linear filters which has the competence to amend the variations of image positioning and it provides information regarding signature using image orientation. It can be computed by following equation

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$g'(x,y) = s'(x,y)w'(x,y)$ (2)

where $s'(x,y)$ is the complex sinusoidal signal carrier and
$w'(x,y)$ is the 2-D gaussian shaped function.

$s'(x,y) = \exp(j(2\pi(u_0 x - v_0 y) + P'$ (3)

Where u_0, v_0 *and* P'defines spatial occurrence and segment of sinusoidal wave respectively.

$w'(x,y) = k \exp(-\pi f)$ (4)

Where f is the highest value of the function and a, b are the scaling parameter of gaussian function.

The main advantage of this feature is that it is scale and rotation invariant. A matrix which is containing the distribution of the co-occurrence values at a given offset over the gray level image is named as Gray- Level Co-occurrence Matrix (GLCM). Co-occurring pair of pixels in GLCM can be spatially related concerning distance and angular spatial relationships which is the advantage of statistical descriptors. The textural descriptors related to GLCM are estimated for four paths including 0º, 45º, 90º, 135º degrees. During this process, only 7 Haralick texture features are obtained from each co-occurrence array [22]. For experimentation, we have extracted Angular Second Moment (ASM), contrast, homogeneity, dissimilarity, entropy and correlation texture descriptors from the grayscale image.

B. 1-D Convolutional Neural Network

After extracting the hand-crafted features of signatures, it will pass on the one-dimensional deep learning model. The reason of using 1-D CNN model is to capture local patterns and features within sequences. Convolutional layers slide a kernel over the input data, identifying relevant patterns in small segments. This enables the model to recognize distinctive features such as low-level and high-level features regardless of their position in the sequence. Similarly, by combining multiple convolutional layers, 1D CNNs can capture compositional features that arise from the combination of simpler patterns. 1D CNNs inherently possess a scale of translation variety, allowing them to identify samples regardless of their precise location in the input sequence. This property is particularly valuable in the signature where the position of feature can vary.

The detail of the layers of 1-D CNN model is given in Fig. 2. The proposed CNN model is less complex and less computationally intense. The proposed model only has six layers including a flatten, ReLU, batch normalization, dropout and dense, and one convolutional layer.



Fig. 2. One-Dimensional CNN model

C. VGG16

VGG16, a CNN architecture designed by the Visual Geometry Group (VGG), was declared winner of 2014 ILSVRC object identification algorithm [23]. Main impact of VGG model was to confirm that the expansion of network depth can boost the performance of model. VGG-16 develops on the guidelines of AlexNet by utilizing several convolution cores, which widen the intensity of the architecture and reduced total parameters [24].

D. MobileNet

MobileNet [25] is distinguished from other extant models by the reduced number of parameters compared to others. It is composed of depthwise discrete convolutions, which were devised to shrink the computing attempts in the first few levels. Its two layers are depthwise and pointwise convolutions. depthwise convolution refers to the process of filtering the input without adding new features. As an outcome, the technique that produces novel descriptors, pointwise convolution, was incorporated. Consequently, the term depthwise discrete convolution was introduced to characterize the combination of two consecutive layers. It allocates a exclusive filter to every input using convolutions and then constructs a dual allocation of output using 1×1 conv (pointwise). After each convolution, Batch Normalization (BN) and Rectified Linear Unit (ReLU) are applied. We extracted the features from fully-connected layer.

E. Dense Net

Huang et al. introduced the Dense Convolution Neural Network DenseNet. [26] Each layer in DenseNet induces all layers in a feed forward approach. Five ranks comprise a DenseNet unit. The initial 4 layers are dense, followed by a transition layer. Each stratum of this dense unit has a growing rate of (k). In the transition layer, 2×2 average pooling layer with a 1×1 convolution and a 2 stride is attached to increase the growth rate to 4. In the dense stratum, there are 1×1 and 3×3 convolutions having a stride of 1. DenseNet169 is utilized in our investigation. We extracted the features from fully connected layer.

F. Bagged Ensamble learning

Ensemble learning describes a broad, effective method to Machine Learning, that shoots to improve analytical capacities by merging the forecasts from numerous simulations. While there are several ensembles to verify signatures, boosting, stacking, and bagging are the three approaches that direct the ensemble learning approach. In this paper, we are using bagging technique. Bagging leads to averaging the verification of signatures from several models that have corresponded to different models of the original datasets.

The signature verification through the ensemble members is merged by majority polling. Three classifiers have been used in our experimentation which includes Support Vector Machine (SVM), Multi-Layer Perceptron (MLP) and Extreme Gradient Bossing (XGB). The entire process varies depending on how all dataset models are made to train ensemble members. The dataset is split up into distinct sections for each member. These members are chosen at random.

The operative theory of SVM is to discover a hyperplane (linear or non-linear) that maximizes the margin. Maximizing the margin is equivalent to:

$$minimize\ \frac{1}{2}w^t w + C\left(\sum_{i=1}^{N}\xi_i\right) \quad (5)$$

$$subject\ to\ y_i(w^T x_i - b) + \xi_i - 1 \geq 0,\ \ 1 \leq i$$

$$\xi_i \geq 0 \quad 1 \leq i$$

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XGB is a supervised ML method that expands performance and speed by employing decision trees. It is quicker to previous performances of gradient boosting. Its objective function is:

$$O(\theta) = L(\theta) + \omega(\theta) \quad (6)$$

where, L denotes the loss function.

IV. Experiments

Three human signature databases, i.e., BHSig260, CEDAR [27] and SigComp2011[28], were used to test the performance of the proposed signature verification model. The BHSig260 dataset has 6,240 real signatures and 7,800 ski fake signatures. These signatures were collected from 260 people. The CEDAR dataset [27] comprises of 55 person's signatures. Each person has presented 24 legitimate signature sample and 24 forged samples. The dataset contains 1,320 (55 × 24) real signatures and 1,320 forged signatures. SigComp2011 was published at the ICDAR 2011 conference for the International Signature Verification Competition (SVC). [28] The dataset contains of both Chinese and Dutch authors. The performance measures to validate the proposed model are Accuracy (ACC), False Acceptance Rate (FAR), and False Rejection Rate (FRR) [29]. These experiments are done using affine transformations. In the testing scenarios, the images are rotated to 90 angle and 10% scaled, as shown in Fig. 3.

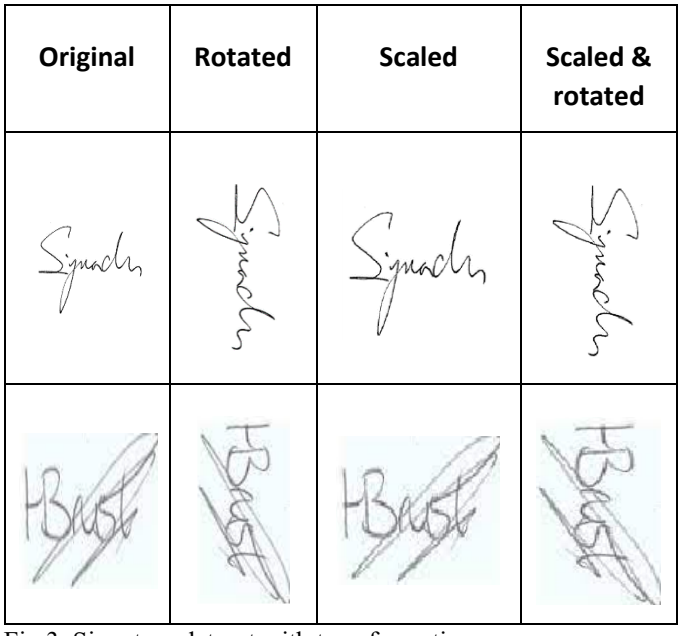
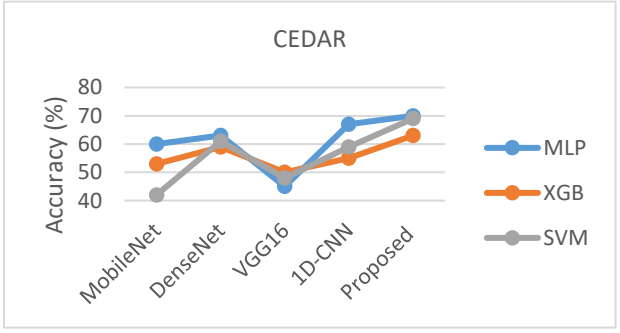
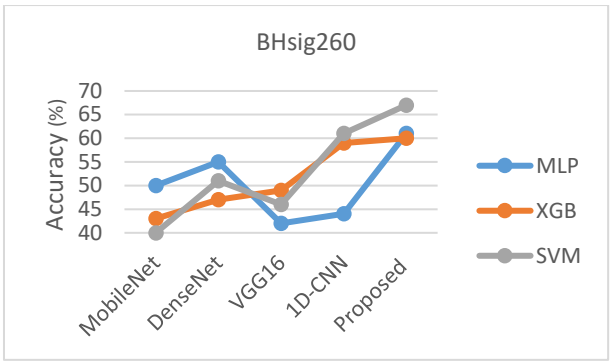


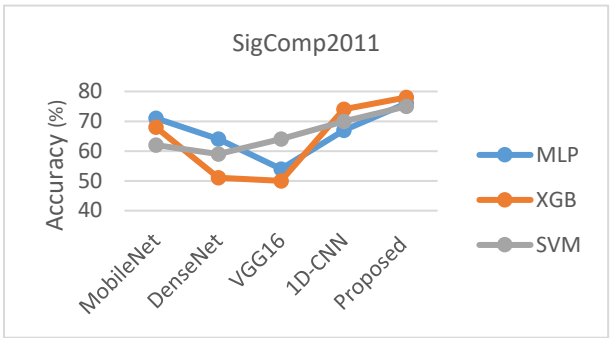
Fig 3. Signature dataset with transformations



(a)



(b)



(c)

Fig. 4 Non-Bagged Ensembled Learning results (a) CEDAR (b) BHsig260 (c) SigComp2011

Fig. 4 demonstrates the non-ensemble learning technique model results using three publicly available datasets. In testing procedure, affine transformations are included. The proposed methodology gave approximately 70%, 68% and 80% on CEDAR, BHsig260 and SigComp2011 datasets respectively. We compared our proposed methodology from the other standalone deep models such as MobileNet, DenseNet, VGG16 and 1-D CNN model and our proposed methodology outperforms among all.

Table I. Results of Bagged Ensemble Learning

Dataset	Perf. Meas.	Bagged Ensemble Learning				
		MobileNet	DenseNet	VGG 16	1D CNN	Proposed Net
CEDAR	FRR	7.8	8.8	5.7	6.3	**4.5**
	FAR	6.1	7.3	6.8	8.2	**3.8**
	ACC	65	73	76	71	**80**
BHsig260	FRR	8.1	9.6	7.9	8.8	**5.1**
	FAR	7.1	8.1	6.1	7.4	**4.7**
	ACC	70	72	63	60	**79**
SigComp 2011	FRR	5.7	6.3	8.7	4.9	**4.7**
	FAR	6.7	5.9	7.4	4.0	**3.6**
	ACC	80	74	81	71	**82**

The results of bagged ensemble learning based on majority polling are given in Table I. The results present the efficiency of this methodology. The proposed methodology is compared with the existing work and its results are given in table II for fair comparison.

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Table II. Comparison of proposed methodology with existing literature

Methods	BHsig260			CEDAR		
	ACC	FAR	FRR	ACC	FAR	FRR
Jagtap et al. [12]	65	9.1	8.8	55	8.4	6.1
Dey et al. [11]	70	8.7	7.4	67	6.1	7.4
Hafemann et al. [13]	62	7.5	6.7	71	5.9	5.6
Proposed	**79**	**4.7**	**5.1**	**80**	**3.8**	**4.5**

The Comparison with the existing techniques is done using the transformed datasets. The proposed methodology has given the promising results as compared to the previous technique.

The proposed methodology is further tested by changing the experimental conditions. The various rotated angles with no scaling factor have been tested on all these three datasets and the results are presented in table III.

Table III. Results of rotation transformation

Dataset	Perf. Meas.	Proposed Methodology				
		30º	60º	90º	180º	270º
CEDAR	FRR	4.5	4.1	4.9	5.1	4.2
	FAR	4.2	3.9	5.4	5.6	4.4
	ACC	79.0	81.2	77.3	76.8	75.4
BHsig260	FRR	6.5	6.8	7.2	6.3	7.0
	FAR	6.7	7.1	6.5	5.9	7.3
	ACC	81.1	79.3	77.6	78.4	79.7
SigComp2011	FRR	5.9	5.6	5.0	5.7	6.8
	FAR	5.3	5.1	4.7	5.9	6.1
	ACC	77.9	79.3	81.8	83.2	80.3

By testing the signature samples at various scaling factors, the proposed methodology has been given promising results, as shown in table IV.

Table IV. Results of scaling transformation

Dataset	Perf. Meas.	Proposed Methodology				
		10	20	25	30	35
CEDAR	FRR	4.7	5.4	5.7	5.8	5.2
	FAR	4.8	5.0	5.1	5.5	4.9
	ACC	78.2	71.8	73.9	74.1	73.4
BHsig260	FRR	6.7	6.1	6.2	5.9	6.9
	FAR	6.3	5.9	5.9	5.2	6.6
	ACC	80.2	80.0	78.5	77.1	74.3
SigComp2011	FRR	5.3	6.1	6.5	6.3	7.1
	FAR	5.9	6.3	6.6	5.8	7.3
	ACC	82.6	81.8	78.9	75.2	79.9

The advantage of using 1D-CNN and hand-crafted features is that the signature's location does not matter as well as the affine transformation can be catered efficiently.

V. CONCLUSION

This paper presented a feature fusion approach in which hand-crafted and deep CNN features have been combined for signature verification. The extracted features from VGG16, MobileNet, DenseNet and 1-D CNN have been compared with the proposed feature fusion technique and from results it reveals that it outperforms as compared to the other techniques. The proposed model has been tested using three popular machine learning classifiers i.e., SVM, MLP and XGB. The result of machine learning classifiers has been selected from bagged ensemble learning approach in which majority voting selected. Image transformations such as rotation and scaling have been considered for testing the feasibility of proposed methodology. Through different experimentation, it has been revealed that the proposed methodology outperforms as compared to the existing work if signature has various transformations other than original sample. This technique can be further tested for on-line signatures to analysis the effectiveness of this technique

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